Data Analysis

Megan Bedell

What can your data tell you?

1. data as information carriers

data → science

science ? → $\underbrace{\text{data}}_{\text{information}}$ → science answer

data taking you from:

high uncertainty → lower uncertainty
less information → more information
high entropy → lower entropy

did I get a talk? → → yes / ~~no~~

what's the composition of a star? → → [Fe/H]

what is Ω_{PM}? → → →

not all data carries equal information / is equally helpful to your science. information theory is about quantifying information content.

if we have a generative probabilistic model $p(x \mid \theta)$, we can follow the information flow to see how much info our data should contain.

areas where information theory is useful:

see original work by Shannon

- encoding / decoding information

  e.g. how many bits are needed to transmit a message without loss?

  how much (& how) can data be compressed with what degree of loss?

- noisy signals

  e.g. how much does the received information get altered / degraded by a noisy transmission channel?

  what is the maximum capacity of a noisy channel?

focus of this lesson

- experimental design

  e.g. what data collection strategy will yield the most information about my problem?

  how much data do I need to collect?

- inference & estimation

  e.g. how precisely <u>should</u> my data constrain my parameters?

- thermodynamics

  e.g. what is the energy cost of making, storing, and erasing measurements?

& much more ...

~ math digression! ~

A well-supported, normalized likelihood function has:

$$\int p(x \mid \theta)\, dx = 1$$

$p = \mathcal{L}$ here (sorry)

differentiate w.r.t. θ:

$$\int \frac{d}{d\theta} p(x \mid \theta)\, dx = 0$$

now use $\frac{d}{d\theta} \ln f = \frac{1}{f} \frac{df}{d\theta}$: $\rightarrow$ $\frac{d \ln \mathcal{L}}{d\theta} = \frac{1}{p} \frac{dp}{d\theta}$

$$\int p(x \mid \theta) \frac{d \ln \mathcal{L}}{d\theta}\, dx = 0$$

$$\mathbb{E}_d\left[\frac{d \ln \mathcal{L}}{d\theta}\right] = 0$$

differentiate again w.r.t. θ:

$$\frac{d}{d\theta} \int p(x \mid \theta) \frac{d \ln \mathcal{L}}{d\theta}\, dx = 0$$

$$\int p(x \mid \theta) \left[\frac{d^2 \ln \mathcal{L}}{d\theta^2} + \left(\frac{d \ln \mathcal{L}}{d\theta}\right)^2\right] dx = 0$$

$$\mathbb{E}\left[\frac{d^2 \ln \mathcal{L}}{d\theta^2}\right] = -\mathbb{E}\left[\left(\frac{d \ln \mathcal{L}}{d\theta}\right)^2\right]$$

2. quantifying information: score functions + Fisher information

score function: $s(\theta) \equiv \frac{\partial}{\partial\theta}(\ln \mathcal{L})$

what's the expected average of $s(\theta)$?

$$\mathbb{E}_d\left[s(\theta)\right] = 0$$

intuitive connection to max likelihood estimation:

- at any <u>one</u> data realization, $\frac{\partial \ln \mathcal{L}}{\partial\theta}$ might not be 0 at the true parameter value θ
- but in the limit of all data, $\frac{\partial \ln \mathcal{L}}{\partial\theta} \to 0$ at θ, so MLE $\hat{\theta}$ approaches the true value!

what's the expected variance of $s(\theta)$?

$$\mathrm{var}(s) = \mathbb{E}(s^2) - \cancel{\mathbb{E}(s)}^2 \to 0$$

$$= -\mathbb{E}\left(\frac{\partial^2 \ln \mathcal{L}}{\partial\theta^2}\right)$$

Fisher information: $\mathrm{var}(s(\theta))$

$$\mathcal{I}(\theta) = -\mathbb{E}\left[\frac{\partial^2 \ln \mathcal{L}}{\partial\theta^2}\right] \leftarrow$$

$$\mathcal{I}_{ij} = -\mathbb{E}\left[\underbrace{\frac{\partial^2 \ln \mathcal{L}}{\partial\theta_i \partial\theta_j}}_{\text{Hessian matrix}}\right]$$

Fisher information matters due to the

Cramer - Rao bound:

$$\mathrm{Var}(\hat{\theta}) \geq \frac{1}{\mathcal{I}(\theta)} \quad \text{or} \quad \mathrm{cov}(\hat{\theta}) \geq \underset{\sim}{\mathcal{I}}^{-1}(\theta)$$

so no unbiased estimator can achieve precision any better than the inverse Fisher information.

the one-parameter Gaussian case:

$$\ln \mathcal{L}(\theta) = \sum_{i=0}^{N} \frac{(x_i - \theta)^2}{2\sigma^2} + \text{const.}$$

$$\frac{\partial \ln \mathcal{L}}{\partial \theta} = \frac{x - \theta}{\sigma^2} \cdot N$$

$$\frac{\partial^2 \ln \mathcal{L}}{\partial \theta^2} = -\frac{N}{\sigma^2} \quad \leftarrow \text{data dropped out!}$$

$$\sigma \sim \frac{1}{\sqrt{N}}$$

the multi-dimensional Gaussian case:

$$\ln \mathcal{L}(\vec{\theta}) = -\frac{1}{2}\left[(d-\mu)^T C^{-1} (d-\mu) + N_d \ln 2\pi + \ln \det[C]\right]$$

$$\underset{\sim}{\mathcal{I}}_{i,j} = \left(\frac{\partial \mu}{\partial \theta_i}\right)^T C^{-1} \left(\frac{\partial \mu}{\partial \theta_j}\right) + \frac{1}{2} \mathrm{tr}\left[C^{-1} \frac{\partial C}{\partial \theta_i} C^{-1} \frac{\partial C}{\partial \theta_j}\right]$$

digression: Gupta & Bedell (2024)

https://iopscience.iop.org/article/10.3847/1538-3881/ad4ce6/pdf

Some useful-to-remember properties of Fisher information:

1. in the case of independent data (no correlated noise), Fisher information is additive:

$$\mathcal{I}_{tot} = \sum_{n=0}^{N} \mathcal{I}_n \quad \text{for } N \text{ data points}$$

→ use this to evaluate how information-rich each data point is, or propose new observations

2. in the multi-parameter case,

$$\mathrm{var}(\hat{\theta}_i) \geq \left[\mathcal{I}^{-1}\right]_{ii}$$

→ if there is significant covariance between parameters, $\left[\mathcal{I}^{-1}\right]_{ii} > \mathcal{I}_{ii}^{-1}$

reflecting the lowered precision that comes from marginalizing over nuisance parameters

takeaways so far:

- Fisher information allows us to quantify the best possible precision on parameters given the data
- For many common forms of likelihood functions, the data values drop out, allowing those Fisher info limits to be very easily calculated (e.g. you don't have to simulate data over many trials to know how well your survey will work)

3. given existing data, what is their model-constraining power?

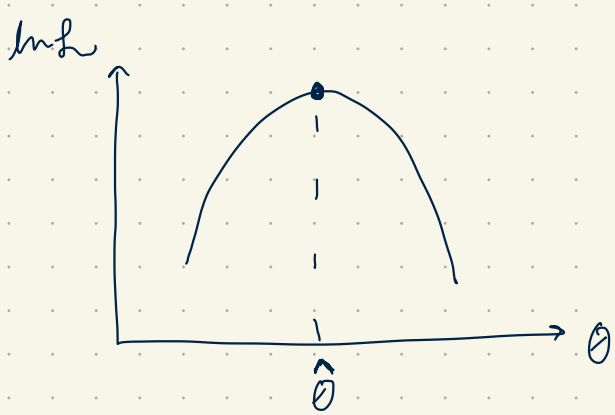
- Fisher information so far has been an expectation over all possible data.
- if your data already exist, you want to instead use the observed Fisher information

$$\hat{\mathcal{I}}_{ij}(\hat{\theta}) = -\left.\frac{\partial^2 \ln \mathcal{L}}{\partial\theta_i \partial\theta_j}\right|_{\hat{\theta}}$$

this is a single-realization sample of the expected Fisher information.

- observed Fisher information is useful when estimating parameter uncertainties from a model fit.

how?



Laplace approximation

$$\ln\mathcal{L}(\theta) = \ln\mathcal{L}(\hat{\theta}) + \cancel{\left.\frac{\partial \ln\mathcal{L}}{\partial\theta}\right|_{\hat{\theta}}}^{\,0}(\theta-\hat{\theta}) + \frac{1}{2}\underbrace{\left.\frac{\partial^2 \ln\mathcal{L}}{\partial\theta^2}\right|_{\hat{\theta}}}_{-\hat{\mathcal{I}}}(\theta-\hat{\theta})^2 + \dots$$

$$\mathcal{L} = p(\theta|x) \propto \exp\left(-\frac{1}{2}(\theta-\hat{\theta})\,\hat{\mathcal{I}}\,(\theta-\hat{\theta})\right)$$

$$\sigma_\theta \sim \hat{\mathcal{I}}^{-1/2}$$

- evaluating the local Hessian around the MLE gives an estimated parameter error
  → this is what many optimizer codes do!

4. empirical alternatives

The Fisher information approach relies on assumptions, including:

- you have a generative model with a twice-differentiable likelihood function supported equally at all θ
- your model is correct (including noise model)
- the likelihood has a single, well-defined peak
- you have plenty of data
- parameters aren't too degenerate

etc etc.

If Fisher info isn't an option, consider injection-recovery testing

draw $\theta_i \sim p(\theta)$

$\downarrow$

generate data $x \sim f(\theta_i, \text{noise})$

$\downarrow$

run analysis + record $\hat{\theta}_i$

repeat many trials

$\Downarrow$

$\{\hat{\theta}_i\}$

Fisher information was a theoretical bound (can't do better), injection-recovery also tests how well your analysis pipeline meets that bound.